SyntheticData.jl - A Julia package to generate autocorrelated synthetic data

Antonino D'Anna

1 Introduction

The package `SyntheticData.jl` provides the tools to generate autocorrelated Monte Carlo data with correlationa and autocorrelations completely under control. The data are generated by repeatedly drawing number from (Multivariate) Normal Distributions and the correlation in Monte Carlo time applied as explained in Sec. 3

2 Multivariate Normal Distribution

In this section we fix the notation and cover the main feature of (Multivariate) Normal Distributions and cover the strategy implemented in `gen_data` and `gen_cov_data`. We begin with a Normal Distribution, whose Probability Distribution Function (PDF) is

$$\mathcal{N}(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \tag{1}$$

where μ is the *true* mean and σ^2 the *true* variance of the distribution. We define also the *Standard* Normal Distribution, as $\mathcal{N}(x, 0, 1)$, which is present in the standard library of any programming language. In particular, the method `randn()` in `Julia`

Let's draw N independent measurements x_i, $i = 1, \ldots, N$, from the distribution, then the observed average is

$$\overline{x} = \frac{1}{N} \sum_{i=1}^{N} x_i \tag{2}$$

and the observed variance $\overline{\sigma}^2$ is

$$\overline{\sigma}^2 = \frac{1}{N-1} \sum_{i=1}^{N} (x_i - \overline{x})^2 \tag{3}$$

In particular, as $N \to \infty$, we have $\overline{x} \to \mu$ and $\overline{\sigma}^2 \to \sigma^2$. For the Normal Distribution, the following theorems are valid:

Theorem 1 *If X is normally distributed according to $\mathcal{N}(x, \mu, \sigma^2)$, then $Y = aX + b$ is normally distributed according to $\mathcal{N}(x, a\mu + b, a^2\sigma^2)$*

x

Theorem 2 *If X_1 is normally distributed according to $\mathcal{N}(x, \mu_1, \sigma_1^2)$ and X_2 is normally distributed according to $\mathcal{N}(x, \mu_2, \sigma_2^2)$, then for any real number a, b, $aX_1 + bX_2$ is normally distributed according to $\mathcal{N}(x, a\mu_1 + b\mu_2, a^2\sigma_2^2 + b^2\sigma_2^2)$.*

Using Th. 1, we can draw number from a Standard Normal Distribution and then covert it to the desire distribution with a linear transformation.

A D-dimensional Multivariate Normal Distribution is defined as

$$\mathcal{N}(\boldsymbol{x}, \boldsymbol{\mu}, \Sigma) = \frac{1}{\sqrt{(2\pi)^D \det(\Sigma)}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \Sigma^{-1} (\boldsymbol{x}-\boldsymbol{\mu})\right), \tag{4}$$

where $\boldsymbol{\mu}$ is a D-dimensional vector that represent the *true* mean vector, and Σ is a $D \times D$ positive definite symmetric matrix that represent the *true* covariance matrix of the distribution. From this distribution, we measure D-dimensional vector $\boldsymbol{x}_i$. The observed mean, $\overline{\boldsymbol{x}}$ is estimated as in eq. (2), while the observed covariance matrix is given by

$$\overline{\Sigma}_{n,m} = \frac{1}{N-1} \sum_{i=1}^{N} (x_{n,i} - \overline{x}_n)(x_{m,i} - \overline{x}_m), \quad n, m = 1, \ldots, D. \tag{5}$$

Once again, in the limit $N \to \infty$, we have $\overline{\boldsymbol{x}} \to \boldsymbol{\mu}$, and $\overline{\Sigma} \to \Sigma$. As for Th. 1, the following theorem is valid:

Theorem 3 *Let $\boldsymbol{Y} = B\boldsymbol{X} + \boldsymbol{b}$, where $\boldsymbol{Y}$ a M-dimensional vector, $\boldsymbol{b}$ is a constant M-dimensional vector, $\boldsymbol{X}$ is a N-dimensional vector and B is a $M \times N$ constant matrix. Then, if $\boldsymbol{X}$ follows $\mathcal{N}(\boldsymbol{x}, \boldsymbol{\mu}, \Sigma)$, then $\boldsymbol{Y}$ follows $\mathcal{N}(\boldsymbol{x}, B\boldsymbol{\mu} + \boldsymbol{b}, B\Sigma B^T)$*

To generate number according to a multivariate distribution, we first generate D independent numbers according to a Standard Normal distribution. Then the desired distribution is obtained by applying the transformation in Th. 3 using a matrix B such that BB^T is the desired covariance matrix.

3 Autocorrelated Data Generation

In this section we cover the generation of autocorrelated data as implemented by the methods `gen_series` and `gen_series!`.

Let's assume we want to generate the autocorrelated Monte Carlo history of length T for a single measurement drawn from a Normal Distribution, eq. (1). Let x_t be the measurement at Monte Carlo time t. Given the observed mean in eq. (2), we define the *deviation*, δ_t, as

$$\delta_t = x_t - \overline{x} \tag{6}$$

By construction, δ_t will we distributed around 0 and it will follow a Standard Normal Distribution. We can define the autocorrelation function, $\Gamma(t - t')$, as the covariance between two measurements taken at Monte Carlo time t and t' respectively. Operatively, we defined the *observed* autocorrelation function as

$$\overline{\Gamma}(t) = \frac{1}{T-t} \sum_{t'=1}^{T-t} \delta_{t'}\, \delta_{t+t'}. \tag{7}$$

In the limit of infinite statistic, is expected to behave as

$$\Gamma(t) \propto \sum_{k=1}^{\infty} \lambda_k^2 e^{-|t|/\tau_k}, \tag{8}$$

where, τ_k is the autocorrelation time and λ_k is the amplitude associated to the $k-$th autocorrelation mode. From eq. (7), we have $\overline{\Gamma}(0) = \overline{\sigma}^2$.

Let assume, for simplicity, that we want to generate a Monte Carlo series with $\Gamma(t)$ defined by one single autocorrelation mode. Using Th. 1, we can draw T random number, ν_t, distributed according to the Standard Normal Distribution $\mathcal{N}(x, 0, \sigma^2)$, with σ^2 the desired variance. Then, the deviation δ_t are constructed as ($a = \exp(-1/\tau)$)

$$\begin{aligned} \delta_1 =& \nu_1 \\ \delta_t =& \sqrt{1-a^2}\,\nu_t + a\,\delta_{t-1} \end{aligned} \tag{9}$$

Then, by defining $x_t = \delta_t + \mu$, we have obtained the desired Monte Carlo chain. Using Th. 2, we can show that, since ν_t follows $\mathcal{N}(x, 0, \sigma)$, also δ_t follows $\mathcal{N}(x, 0, \sigma)$. The correlation in Monte Carlo time constructed "by hand" by eqs. (9). To generalized to the case with N_τ autocorrelation modes, we simply generate for each mode a Monte Carlo series as above, then we obtain the deviation δ_t as

$$\delta_t = \sum_k \lambda_k \delta_{tk} \tag{10}$$

where δ_{tk} is the deviation associated to the $k-$th mode at Monte Carlo time t. Finally, $x_t = \delta_t + \mu$. Using once again Th. 2, we can show that x_t is normally distributed according to $\mathcal{N}(x, \mu, \sigma^2 \sum_k \lambda_k^2)$, hence, it is advisable to always normalize the mode amplitude such that $\sum_k \lambda_k^2 = 1$

Using the method `gen_series`, we generate three data point setting $\mu = 1.0$ and $\sigma^2 = 0.01$, and varying τ. In Fig. 1 we show the normalized autocorrelation function, $\rho(t) = \Gamma(t)/\Gamma(0)$, of the data generated at $\tau = 5, 10, 25$, together with theoretical behavior. We observed that the generated data remarkably agree with the expected behavior. The small deviation observed, for example for $\tau = 40$ are explained by statistical fluctuation. In the legend of Fig. 1, we quote the observed average and the error computed with the Gamma Method [ref]. In Fig. 2, instead, we show an example of normalized autocorrelation function with three autocorrelation modes. For simplicity, we chose three autocorrelation modes with the same amplitude $\lambda_k = 1/\sqrt{3}$ and autocorrelation time of $\tau = 10, 15, 19$. Even in this more complicated case, we manage to recover the expected behavior for $\rho(t)$, with the discrepancies observe typically smaller than two times the uncertainty on the autocorrelation function.

4 Correlated - Autocorrelated data generation

In this section we cover the generation of autocorrelated data as implemented by the method `gen_cov_series`.

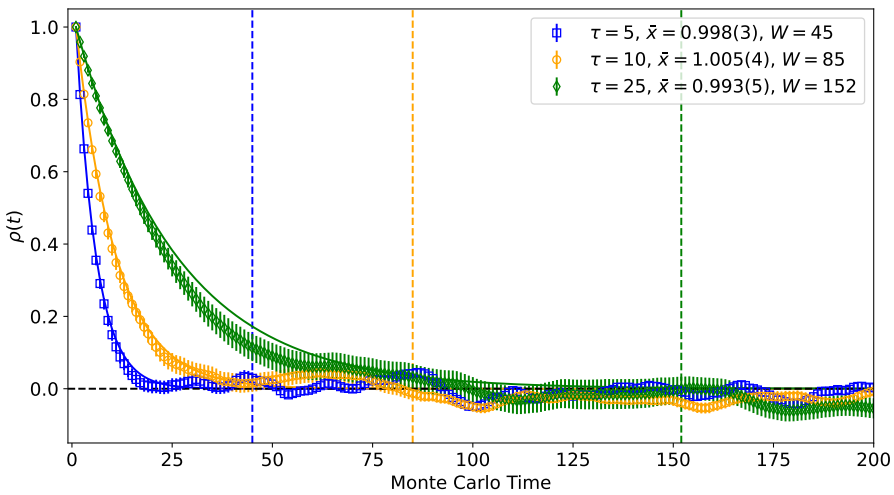

Figure 1: Normalized autocorrelation function $\rho(t) = \Gamma(t)/\Gamma(0)$. The error-bars shown are computed using [ref]. The colored solid line are the theoretical value of $\rho(t)$. In the legend we quote the observed average and the associated error computed using the Gamma Method as outlined in [ref]. With W, we indicate the integration window computed as in [ref] with $S_\tau = 2.0$, also plotted as a vertical dashed line. The data shown where computed on a Monte Carlo Chain of length T = 15000, however we only show $\rho(t)$ for Monte Carlo time $t \leq 200$ for readability.

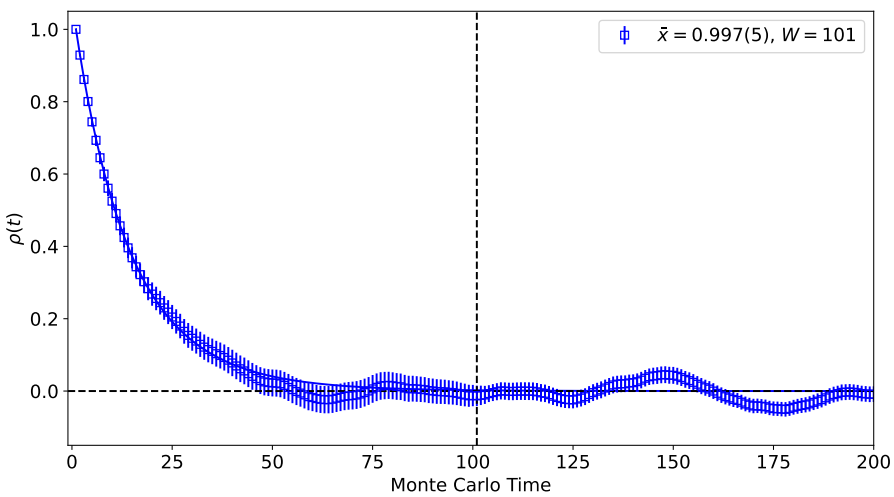

Figure 2: Normalized autocorrelation function $\rho(t) = \Gamma(t)/\Gamma(0)$ computed using three autocorrelation modes corresponding to $\tau = 10, 15, 19$, all with the same amplitude $\lambda = 1/\sqrt{3}$. The error-bars shown are computed using [ref]. The colored solid line represents the theoretical value of $\rho(t)$. In the legend we quote the observed average and the associated error computed using the Gamma Method as outlined in [ref]. With W, we indicate the integration window computed as in [ref] with $S_\tau = 2.0$, also plotted as a vertical dashed line. The data shown where computed on a Monte Carlo Chain of length T = 10000, however we only show $\rho(t)$ for Monte Carlo time $t \leq 200$ for readability.

Let's assume we want to generate a vector $\boldsymbol{x}$ of D correlated data according to the multivariate distribution in eq. (4). By using a Cholesky decomposition, we can write the covariance matrix as

$$\Sigma = LL^T \tag{11}$$

where L a lower triangular matrix. Then, we generate a $D-$vector, $\boldsymbol{\eta}$, of independent random number according to a Standard Normal Distribution and Monte Carlo history as outlined in Sec. 3, then, *for each* Monte Carlo time, the data are given by

$$\boldsymbol{x}_t = L\boldsymbol{\eta}_t + \boldsymbol{\mu} \tag{12}$$

5 Methods

In this section we document all the method exported by `SynteticData.jl`.

- `gen_data(mu, sigma; N=1000, data=false)`
  It draws `N` random numbers from the Normal Distribution $\mathcal{N}(x, \texttt{mu}, \texttt{sigma}^2)$. It returns the observed mean and variance. If `data` is set to `true`,it returns the drawn data.

- `gen_cov_data(mu, cov; data = false, N=1000)`
  It draws `N` vectors of length `length(mu)` from the Multivariate Normal Distribution $\mathcal{N}(x, \texttt{mu}, \texttt{cov})$. It returns the observed mean and covariance matrix. If `data` is set to `true`, it returns the drawn data.

- `random_walk(mean; sigma=1.0, N = 1000)`
  It generates a random walk around `mean` with tolerance `tol` and length `N`. The random numbers are drawn from the Normal Distribution $\mathcal{N}(x, \texttt{mean}, \texttt{tol}^2)$. It returns a vector with the generated data.

- `gen_series!(dt,tau[,lambda]; N=1000, sigma=1.0)`
  It generates a Monte Carlo series of length `N`, with autocorrelations defined by `tau` and `lambda`. The resulting series is stored in `dt`. If `lambda` is not given, then $\texttt{lambda} = 1/\sqrt{\texttt{length(tau)}}$ is assumed

- `gen_series(mu, tau[, lambda]; N=1000, sigma)`
  It generates a Monte Carlo Series of length `N` with autocorrelations defined by `tau` and `lambda`. If `lambda` is not given, then $\texttt{lambda} = 1/\sqrt{\texttt{length(tau)}}$ is assumed. It returns the Monte Carlo series generated

- `gen_cov_series(mu, cov, tau[, lambda]; N=1000)`
  It generates `length(mu)` Monte Carlo series of length `N` with autocorrelations defined by `tau` and `lambda` and covariance defined by `cov`. If `lambda` is not given, then $\texttt{lambda} = 1/\sqrt{\texttt{length(tau)}}$ is assumed. It returns the Monte Carlo series generated.

- `real_taui(tau [,lambda])`
  It returns the true $\tau_{\rm int}$ associate to the autocorrelation function defined by `tau` and `lambda`. If `lambda` is not given, then `lambda` $= 1/\sqrt{\texttt{length(tau)}}$ is assumed.

- `to_cov(corr,sigma)`
  It returns the covariance matrix given the correlation matrix `corr` and the standard deviations `sigma`

- `to_corr(cov)`
  It returns the correlation matrix associated to the covariance matrix `cov`